

Step-by-Step Tutorial

Grain-Size Analysis with the Sedimentology-GrainSize toolkit

1. What this toolkit does

Given a table of grain-size measurements, the toolkit runs seven standard sedimentological analyses and assembles the results into publication-ready figures and reports. The seven analyses are:

- 1 Grain-size histogram (frequency per class)
- 2 Smooth frequency distribution curve with mode detection
- 3 Cumulative curve and Folk & Ward (1957) statistics
- 4 Passega C-M transport analysis
- 5 Folk (1954) ternary classification
- 6 Shepard (1954) ternary classification
- 7 Visher (1969) log-probability transport-population analysis

2. Preparing your input CSV

The WIDE format (recommended): the first column is the sample name; every other column header is a grain size class value; each cell is the weight percent in that class.

```
sample, -1, -0.5, 0, 0.5, 1, 1.5, 2, 2.5, 3, ...  
RIV-01, 0.0, 0.2, 1.1, 3.5, 8.2, 14.1, 18.6, 19.0, 15.3, ...
```

Column headers may be in phi, mm, or micrometres - set **size_unit** accordingly. Rows need not sum to 100; they are normalised automatically. A LONG (tidy) layout with sample/size/value columns is also accepted and auto-detected.

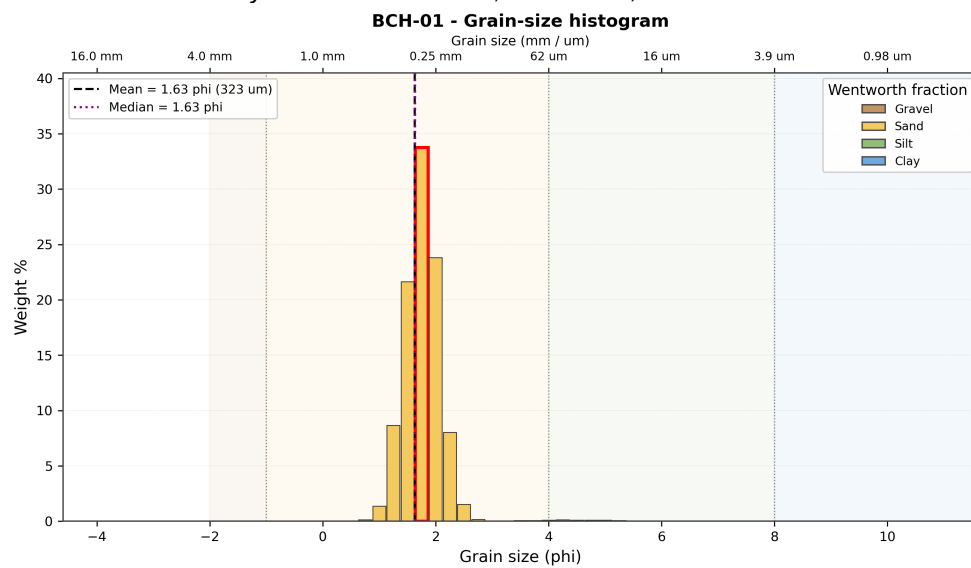
3. Running the analysis

Open **notebooks/Master_Workflow.ipynb** (offline) or **Master_Workflow_Colab.ipynb** (Colab), then run the cells top to bottom. Or from a Python session:

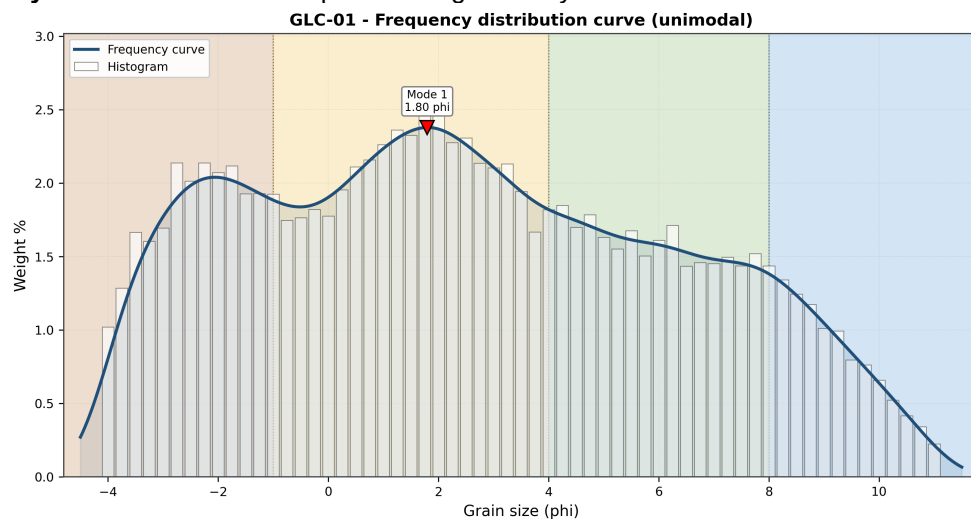
```
import grainsize as gs  
out = gs.run_workflow('sample_data/beach.csv', outdir='output')
```

4. Reading the outputs

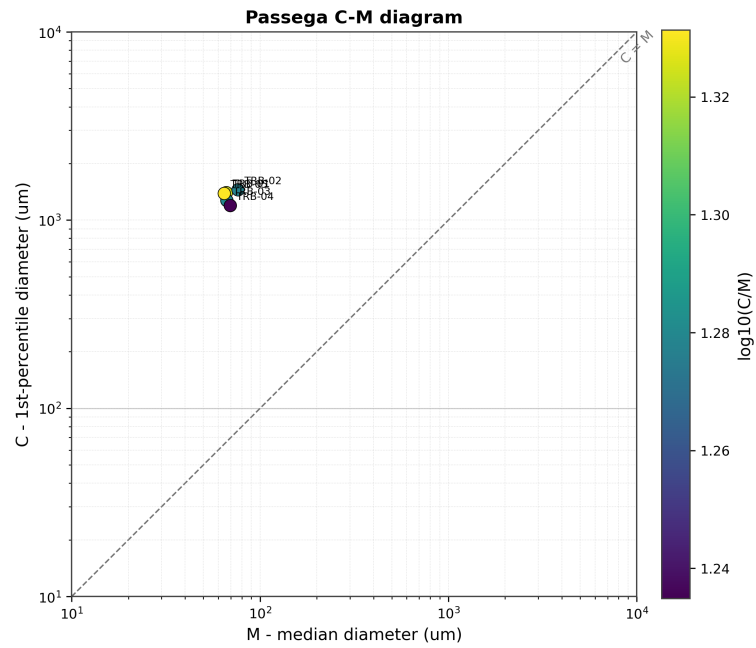
Histogram - bars coloured by Wentworth fraction, with mean, median and modal class marked:



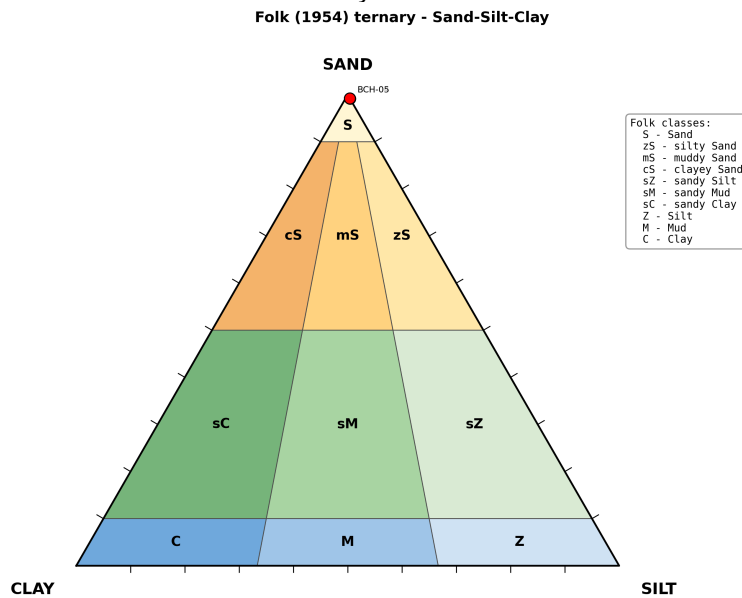
Frequency curve - smooth envelope revealing modality:



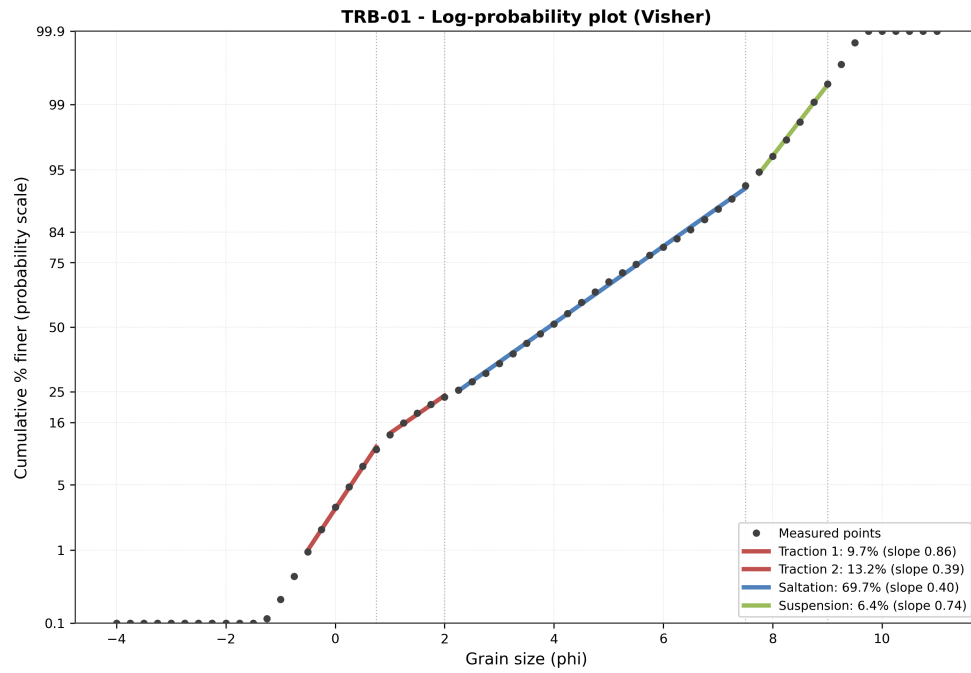
Passega C-M diagram - transport mechanism from the coarsest (C) and median (M) grain sizes:



Folk ternary - textural class from sand-silt-clay ratios:



Visher log-probability plot - traction/saltation/suspension sub-populations detected automatically:



5. The generated reports

Every run produces a professional PDF and a self-contained HTML report with a cover page, per-sample results, all figures, interpretations, and references.

Grain-Size Analysis Report

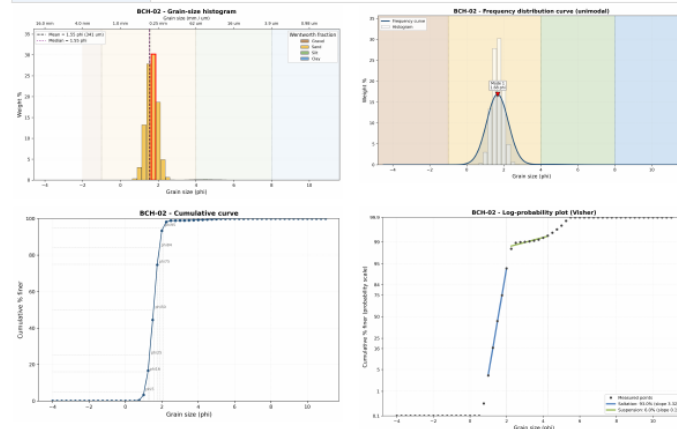
Dataset: **beach**
5 samples • 7 analyses per sample

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Sedimentology-GrainSize v1.0.0

A per-sample page from the PDF report:

BCH-02

Mean	1.55 phi (341 um) - medium sand
Sorting	0.32 - very well sorted
Skewness	0.03 - near-symmetrical
Kurtosis	1.00 - mesokurtic
Modality	unimodal
Folk / Shepard	Sand / Sand
Passega C-M	C=573 um, M=343 um, C/M=1.7
Transport	Uniform suspension / sorted traction
Visher env.	Aeolian dune sand - exceptional saltation sorting



Interpretation: The sample is a sand (Folk 1954; Shepard: sand), with a graphic mean of 1.55 phi (341 um, medium sand) and is very well sorted, near-symmetrical, and mesokurtic. The distribution is unimodal with a primary mode at 1.68 phi. On the Passega C-M diagram the sample plots with C = 573 um and M = 343 um (C/M = 1.7), indicating uniform suspension / sorted traction. Log-probability segmentation suggests: Aeolian dune sand - exceptional saltation sorting.

6. Running a single analysis

If you only need one analysis, open the matching notebook in **notebooks/** (01_Histogram through 07_Visher_Probability). Each is self-contained and follows the same load-analyse-plot pattern.

7. Troubleshooting

If a notebook cannot find the **grainsize** package or **sample_data/**, make sure you run it from within the repository - the setup cell changes the working directory to the repo root automatically. See the README FAQ for more.